GHAZI UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE

PROGRAMMING FUNDAMENTALS

LAB #04

LABORATORY ASSIGNMENT

STUDENT NAME

Muhammad Yaseen

ROLL NUMBER

15

PROGRAM

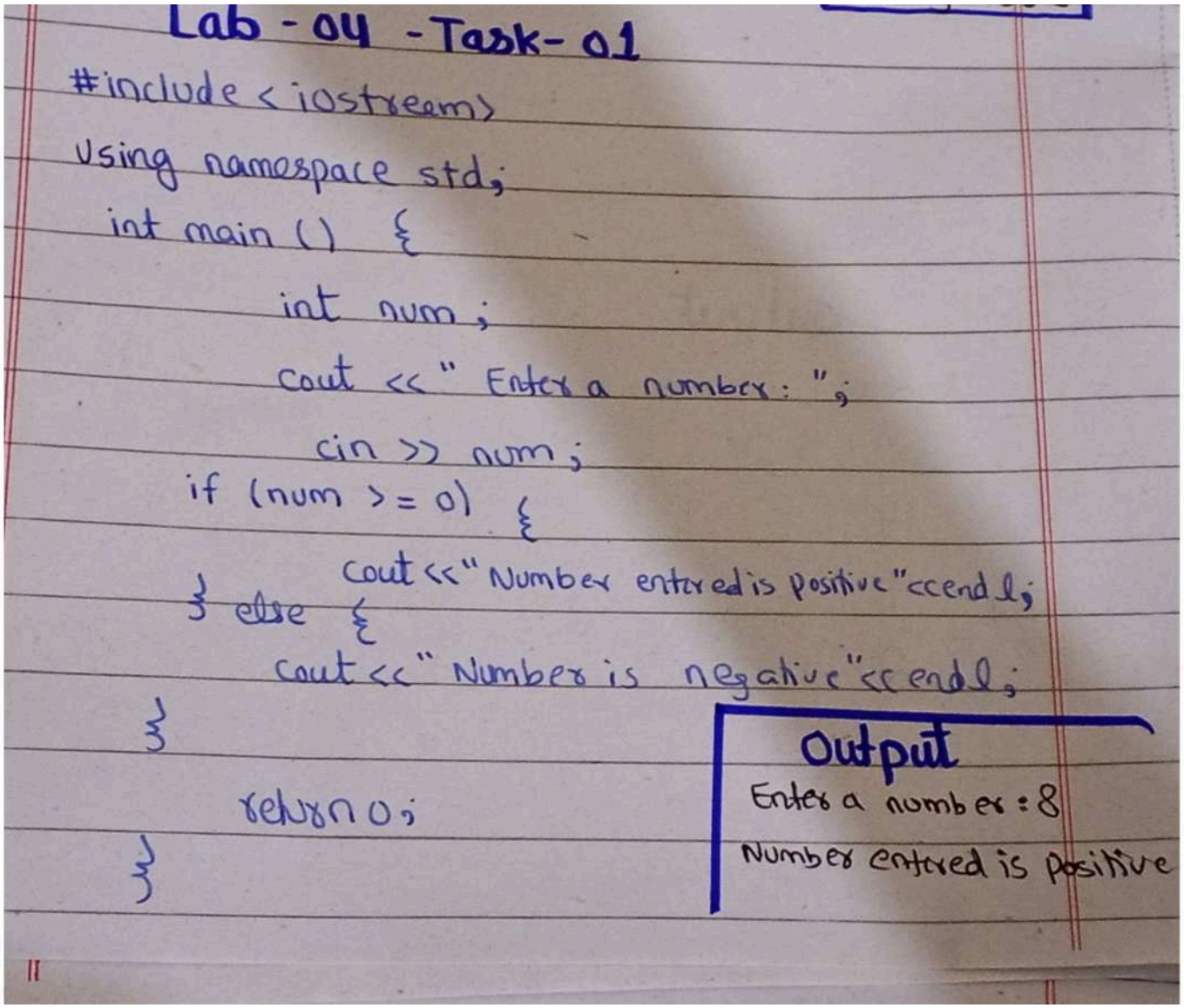
BS Computer Science

SECTION

A

SUBMITTED TO

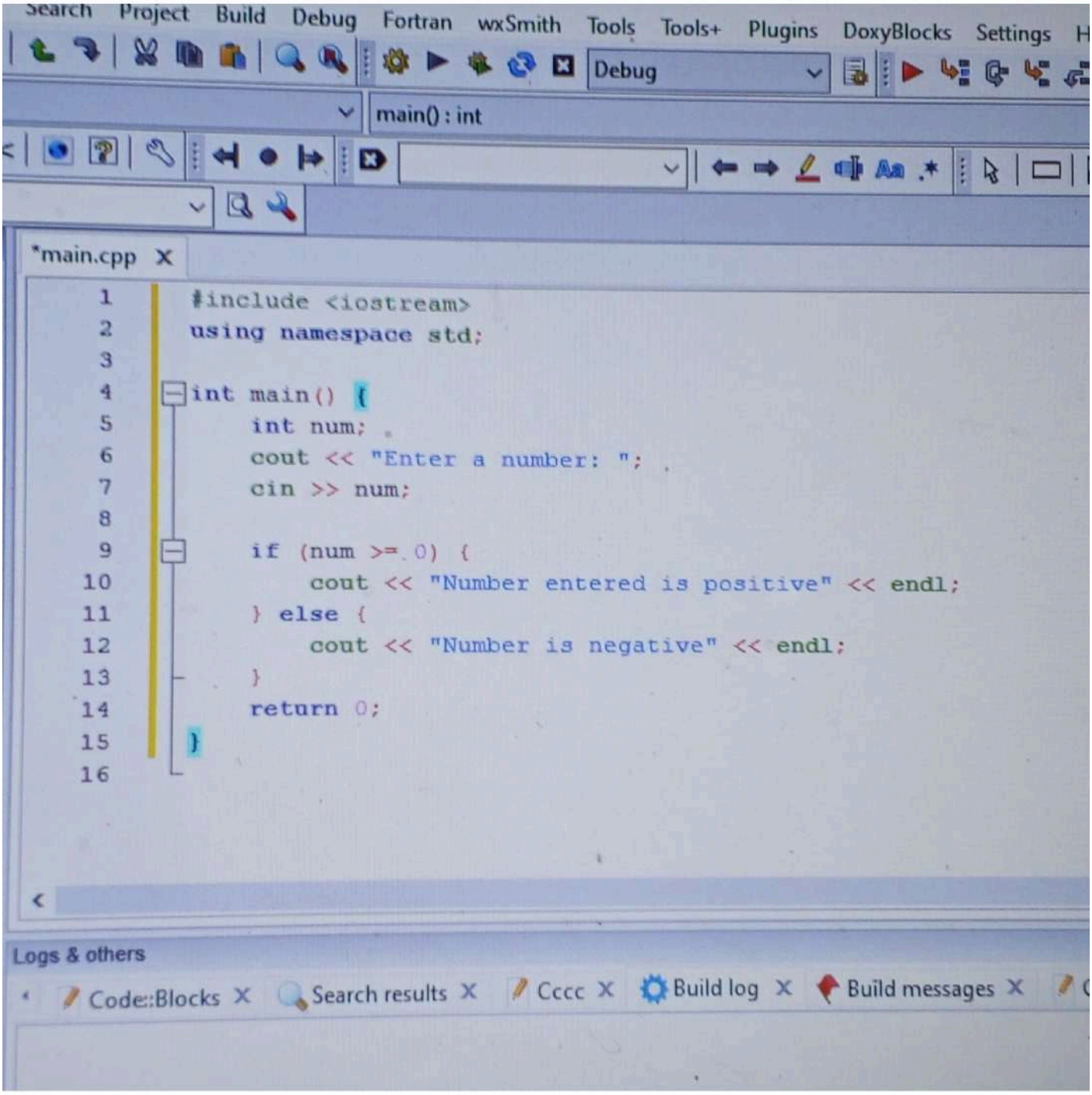
Sir Asad Majeed

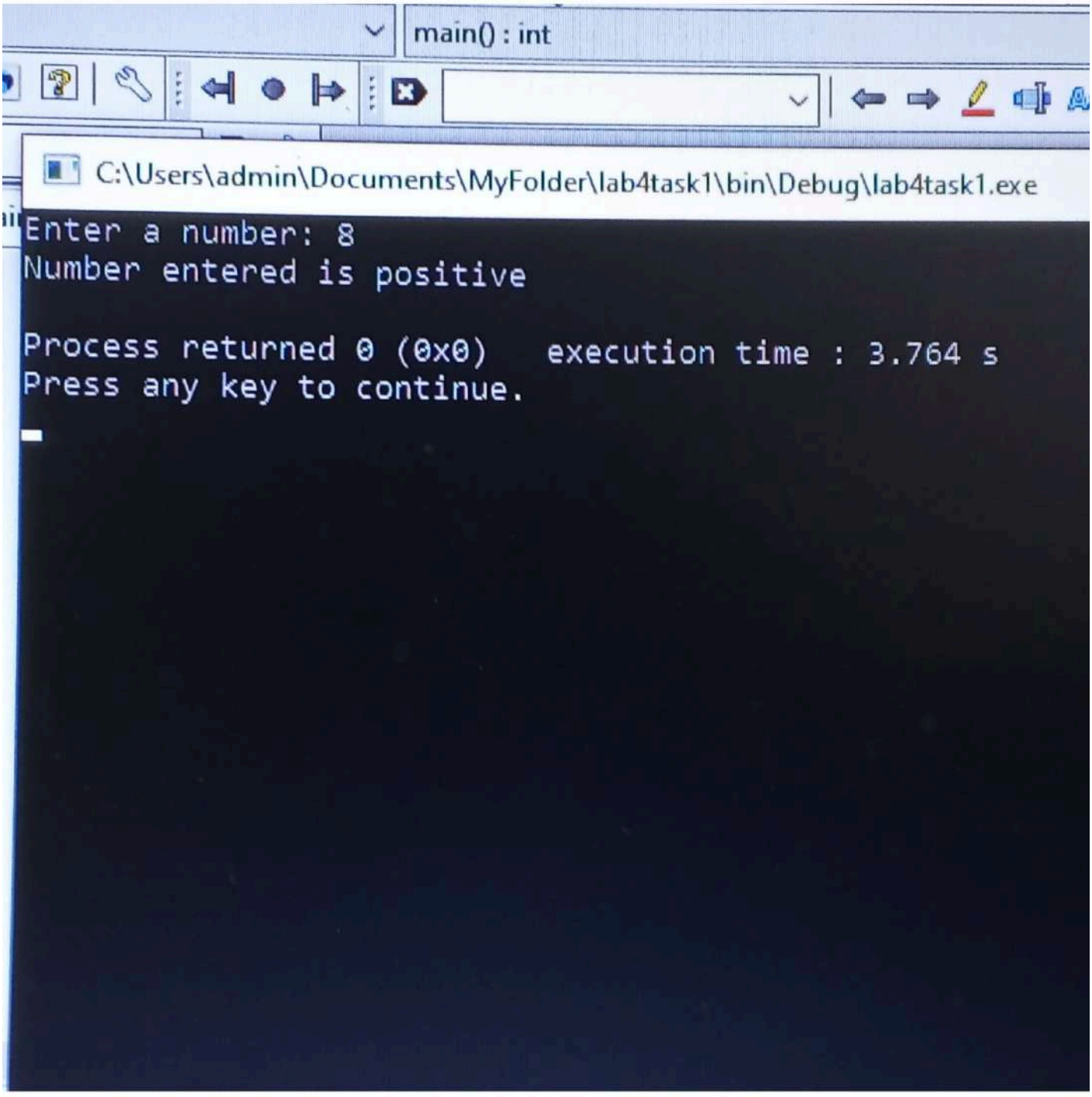


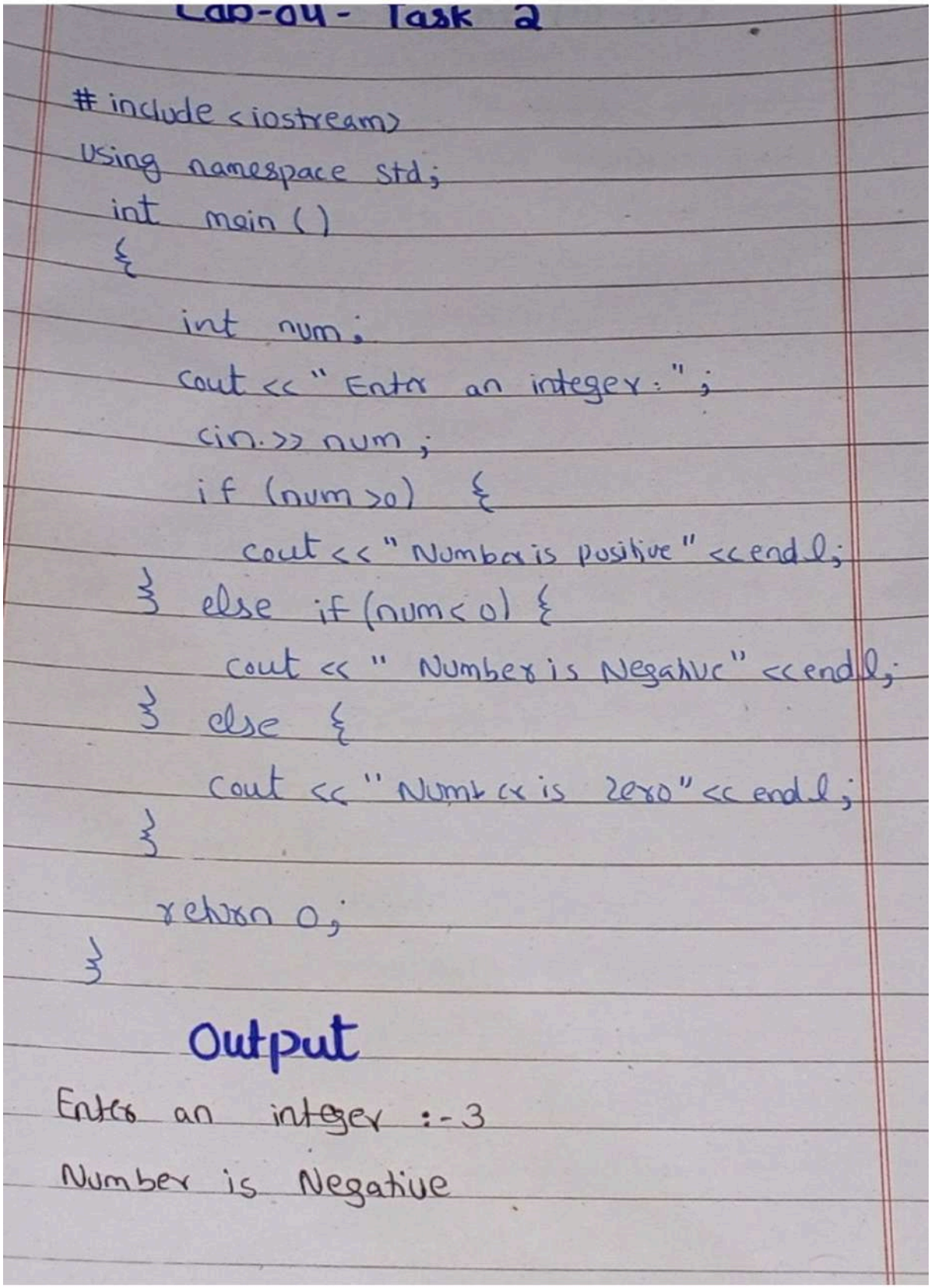
Write a program to print the number entered by the user and determine whether it is positive or negative.

QUESTION

LAB TASK 1 · POSITIVE OR NEGATIVE NUMBER


Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings H
Debug
main() : int
*main.cpp
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5 int num;
6 cout << "Enter a number: ";
7 cin >> num;
8
9 if (num >= 0) {
10 cout << "Number entered is positive" << endl;
11 } else {
12 cout << "Number is negative" << endl;
13 }
14 return 0;
15 }
16
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages


main() : int
C:\Users\admin\Documents\MyFolder\lab4task1\bin\Debug\lab4task1.exe
Enter a number: 8
Number entered is positive
Process returned 0 (0x0)   execution time : 3.764 s
Press any key to continue.



Lab-04 - Task 2

```
#include <iostream>
using namespace std;
int main()
{
    int num;
    cout << "Enter an integer: ";
    cin >> num;
    if (num > 0) {
        cout << "Number is positive" << endl;
    } else if (num < 0) {
        cout << "Number is Negative" << endl;
    } else {
        cout << "Number is zero" << endl;
    }
    return 0;
}
```

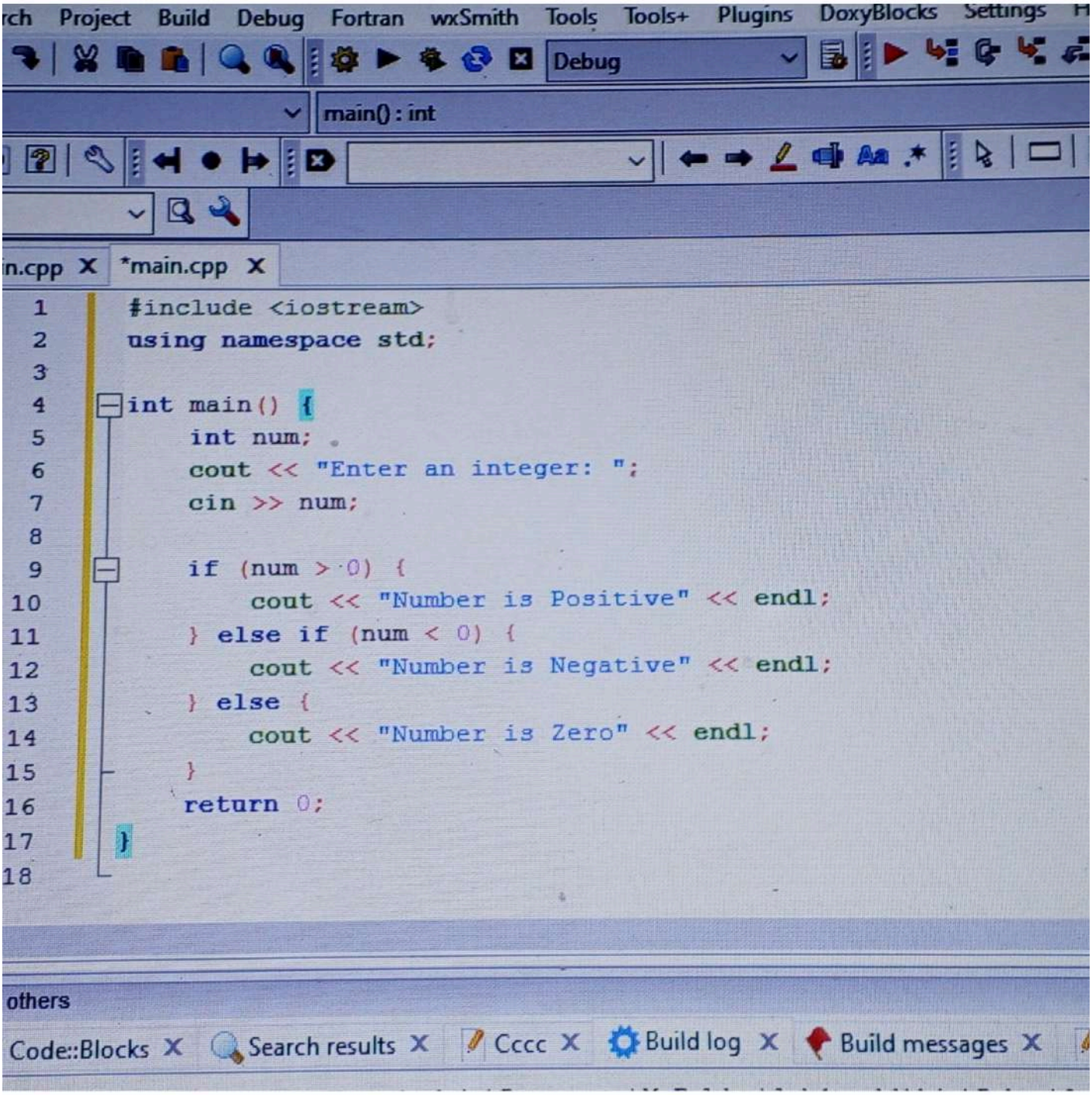
Output

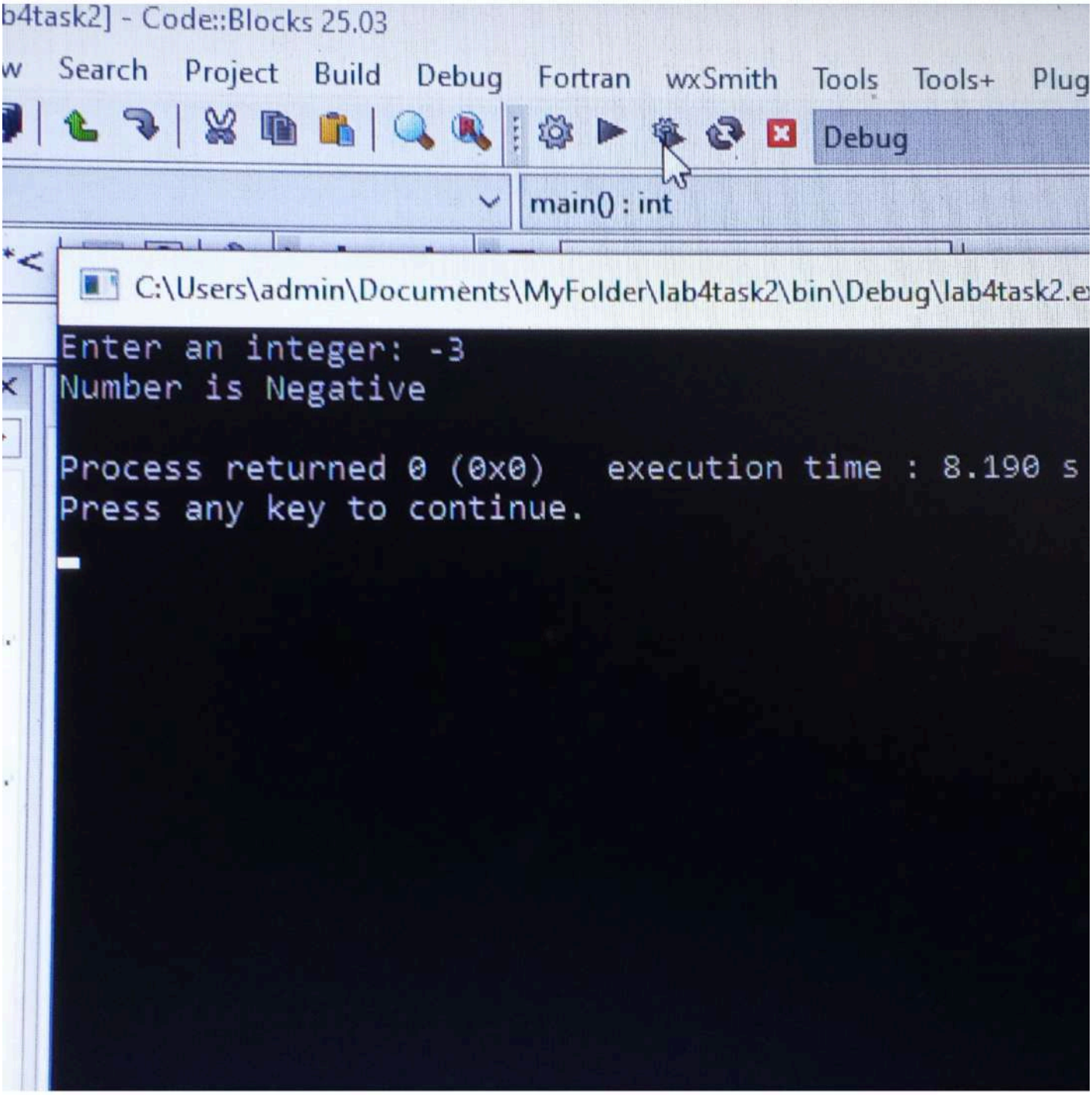
```
Enter an integer :-3
Number is Negative
```

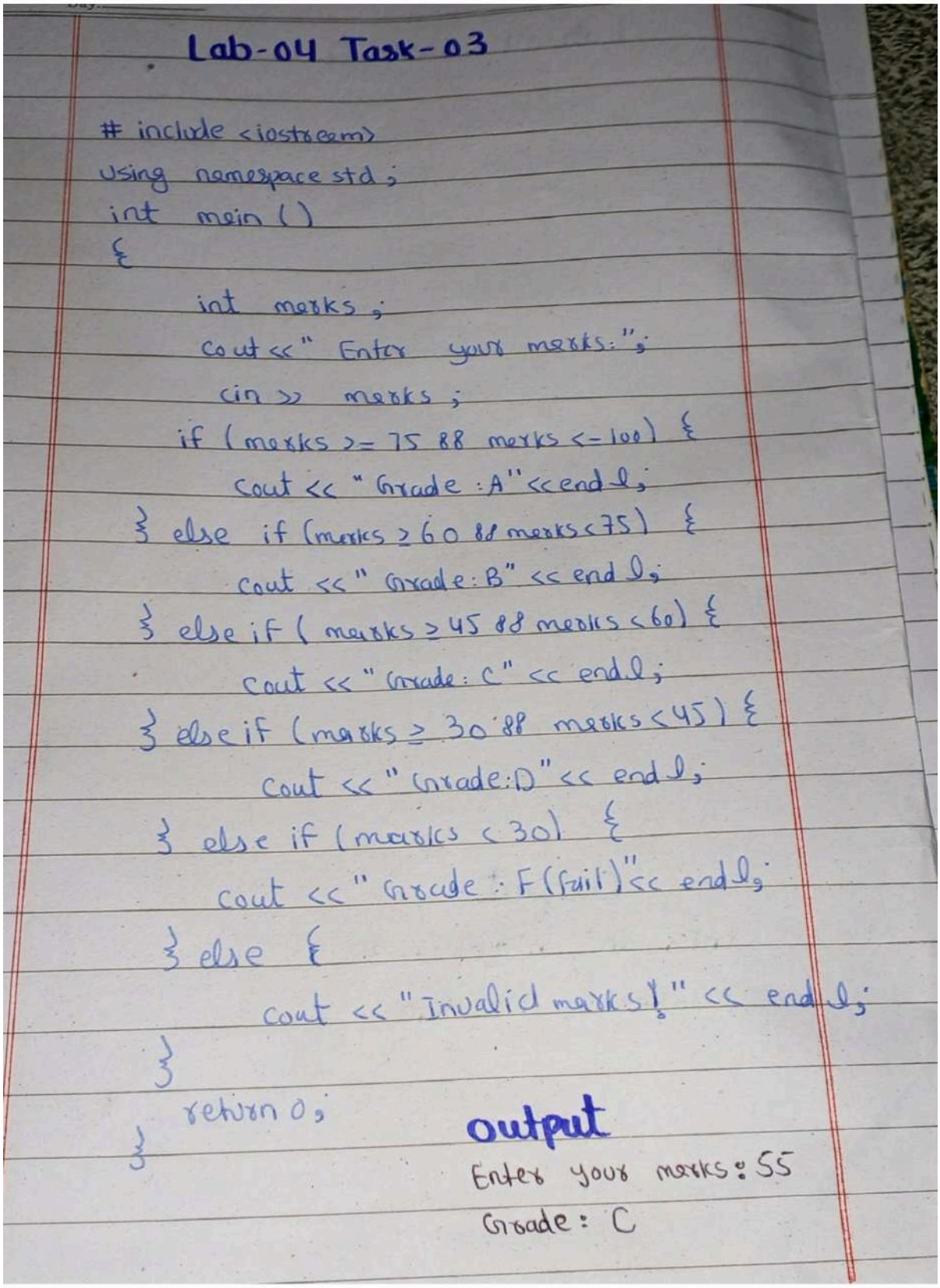
Write a program to check whether an integer is positive, negative, or zero.

QUESTION

LAB TASK 2 · POSITIVE, NEGATIVE OR ZERO


Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings
Debug
main() : int
*main.cpp
#include <iostream>
using namespace std;
int main() {
int num;
cout << "Enter an integer: ";
cin >> num;
if (num > 0) {
cout << "Number is Positive" << endl;
} else if (num < 0) {
cout << "Number is Negative" << endl;
} else {
cout << "Number is Zero" << endl;
}
return 0;
}
others
Code::Blocks
Search results
Cccc
Build log
Build messages

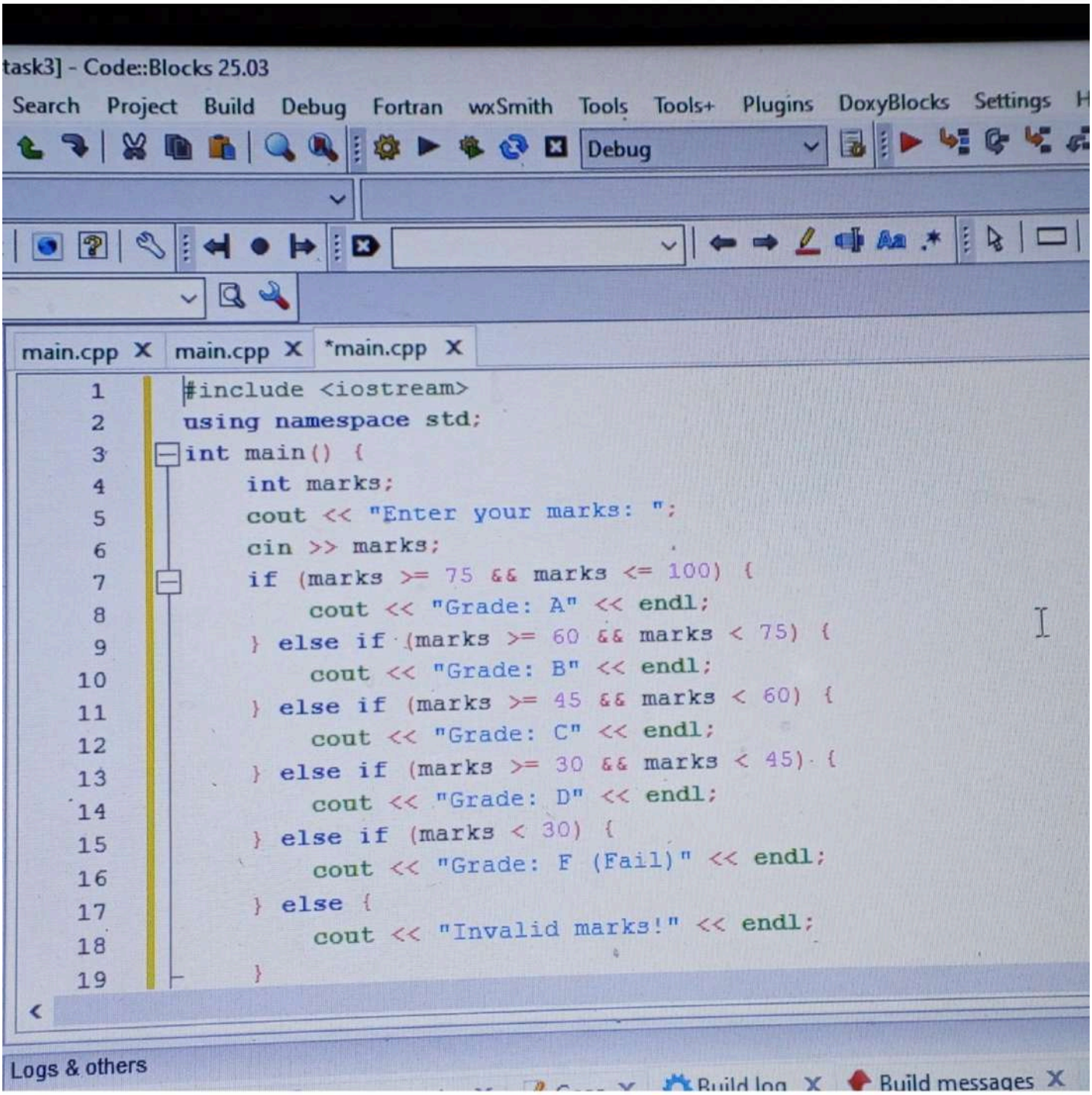

b4task2] - Code::Blocks 25.03
w Search Project Build Debug Fortran wxSmith Tools Tools+ Plug
Debug
main() : int
C:\Users\admin\Documents\MyFolder\lab4task2\bin\Debug\lab4task2.e
Enter an integer: -3
Number is Negative
Process returned 0 (0x0) execution time : 8.190 s
Press any key to continue.

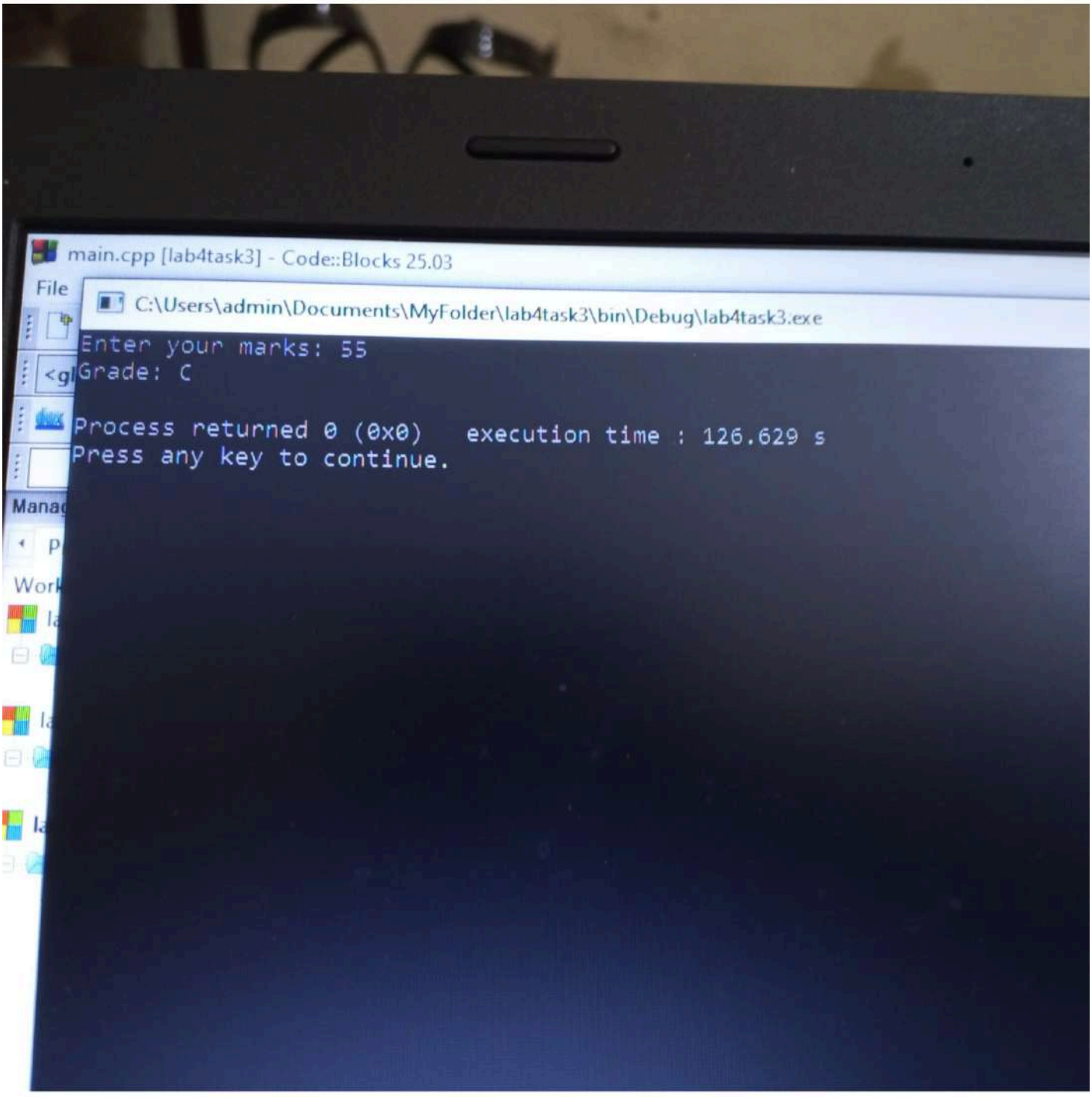


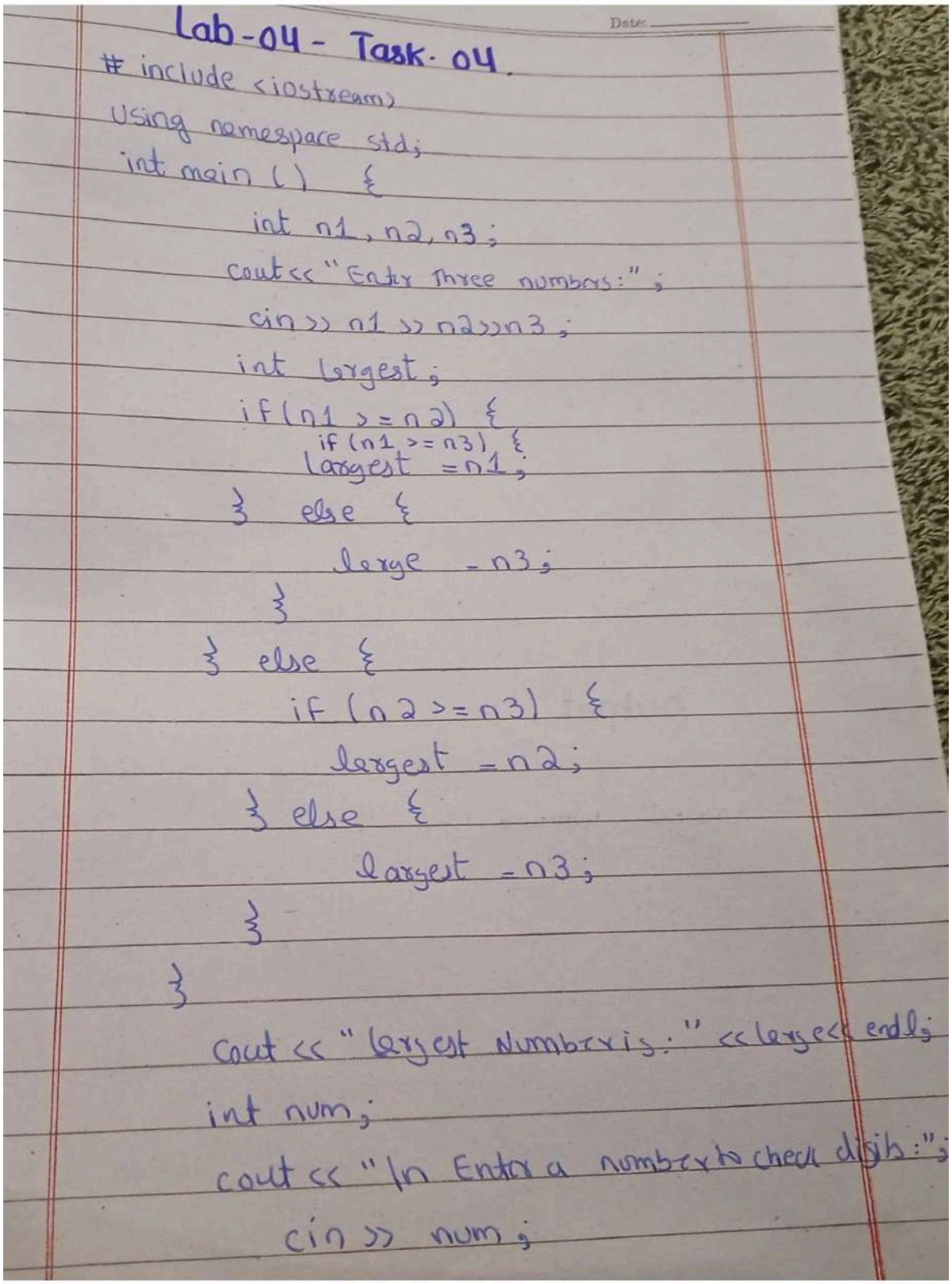
Use the given mark ranges to determine and display the student·s grade.

QUESTION

LAB TASK 3 · MARKS AND GRADING


task3] - Code::Blocks 25.03
Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings
Debug
main.cpp
main.cpp
*main.cpp
#include <iostream>
using namespace std;
int main() {
int marks;
cout << "Enter your marks: ";
cin >> marks;
if (marks >= 75 && marks <= 100) {
cout << "Grade: A" << endl;
} else if (marks >= 60 && marks < 75) {
cout << "Grade: B" << endl;
} else if (marks >= 45 && marks < 60) {
cout << "Grade: C" << endl;
} else if (marks >= 30 && marks < 45) {
cout << "Grade: D" << endl;
} else if (marks < 30) {
cout << "Grade: F (Fail)" << endl;
} else {
cout << "Invalid marks!" << endl;
}
Logs & others
Build log
Build messages


main.cpp [lab4task3] - Code::Blocks 25.03
File
C:\Users\admin\Documents\MyFolder\lab4task3\bin\Debug\lab4task3.exe
Enter your marks: 55
Grade: C
Process returned 0 (0x0)   execution time : 126.629 s
Press any key to continue.



digits.

Find the largest of three numbers using nested if-else statements, then check whether an entered number is positive and count its

QUESTION

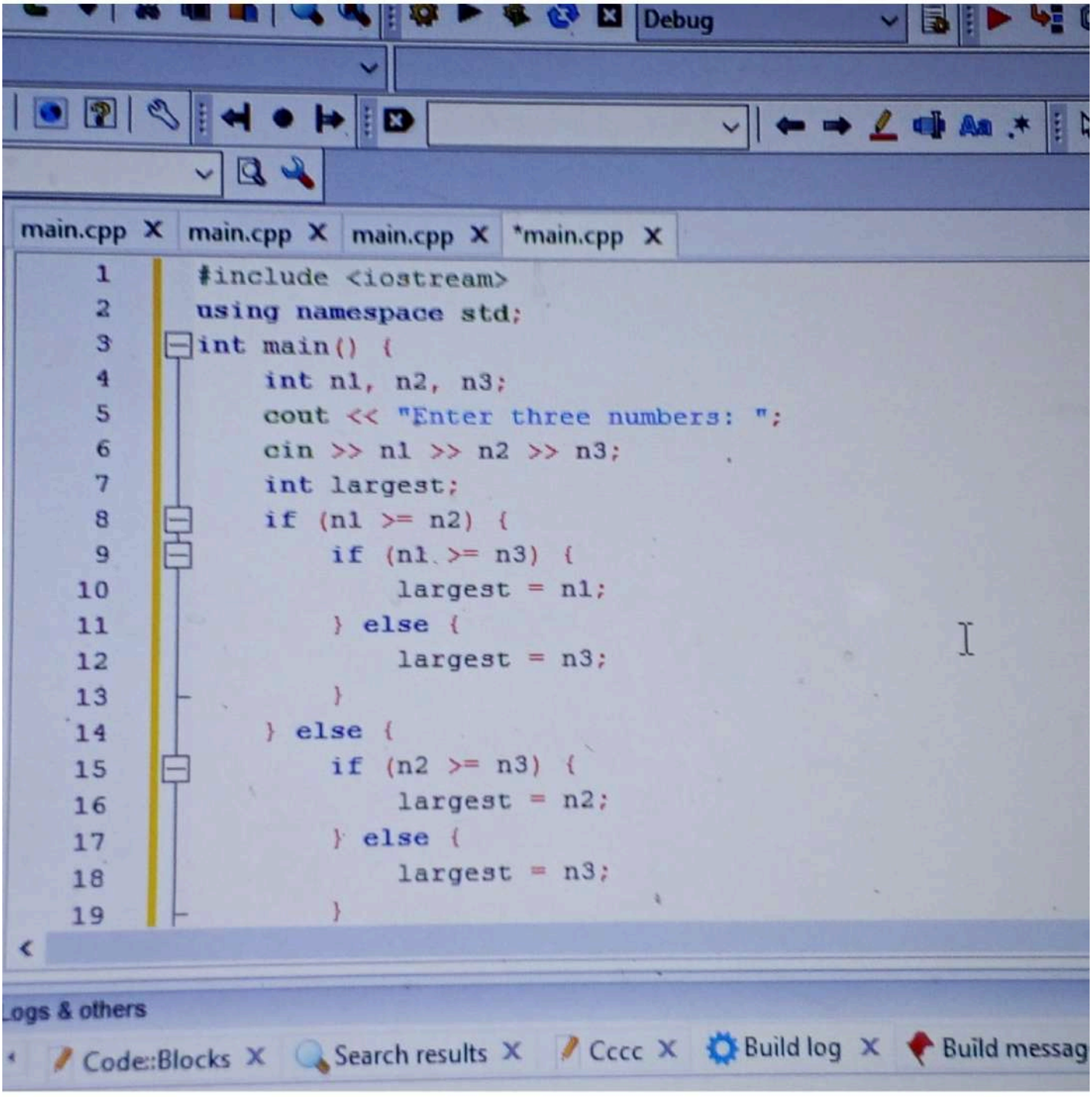
LAB TASK 4 · LARGEST NUMBER AND DIGIT COUNT

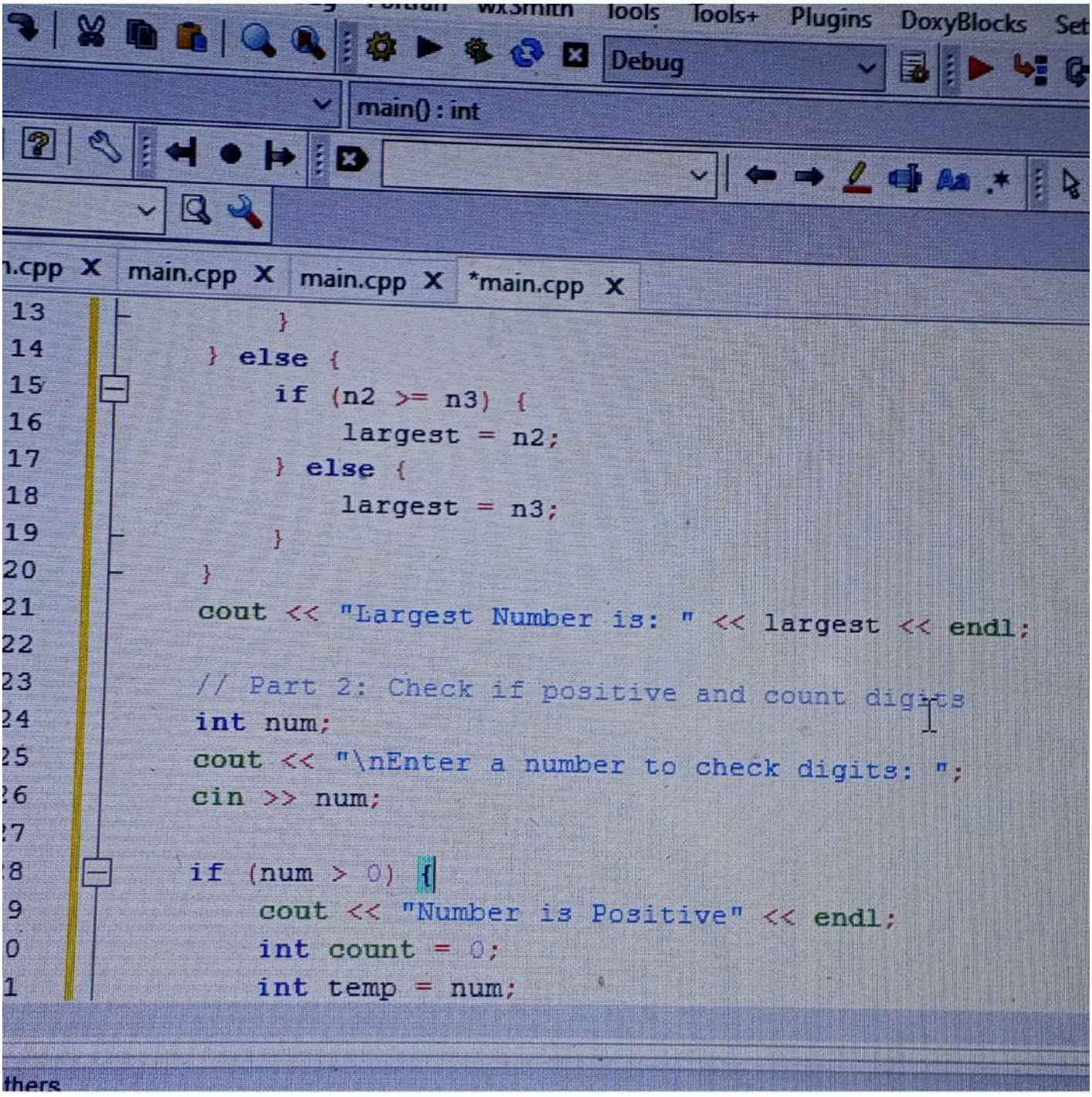
```
if (num >0) {
    cout << "Number is Positive" << endl;
    int count = 0;
    int temp = num;
    while ( temp != 0) {
        temp = temp / 10;
        count ++;
    }
    cout << " Number of digits :" << count << endl;
} else {
    cout << "Number is not positive" << endl;
}
    return ;
}
```

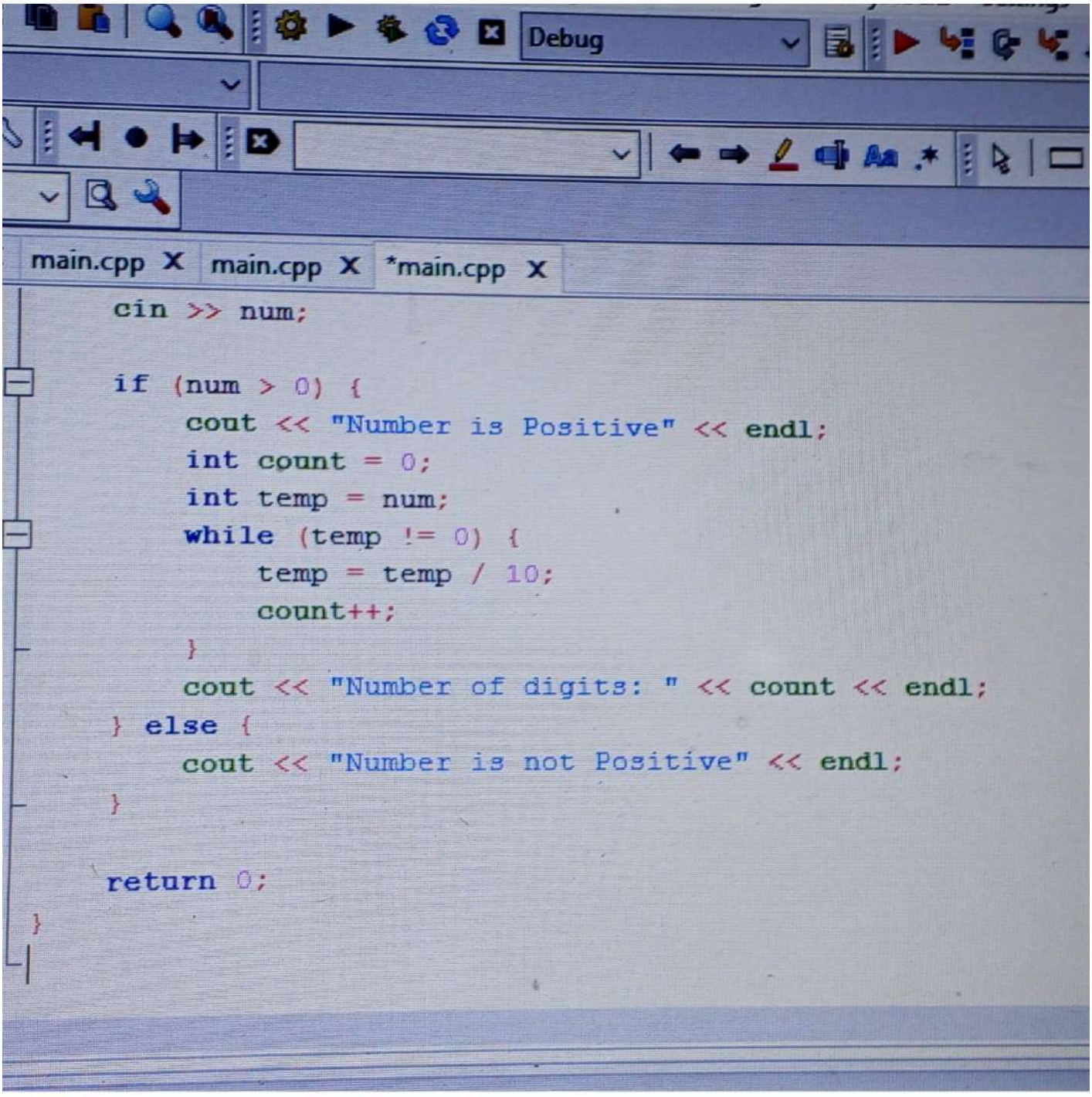
output

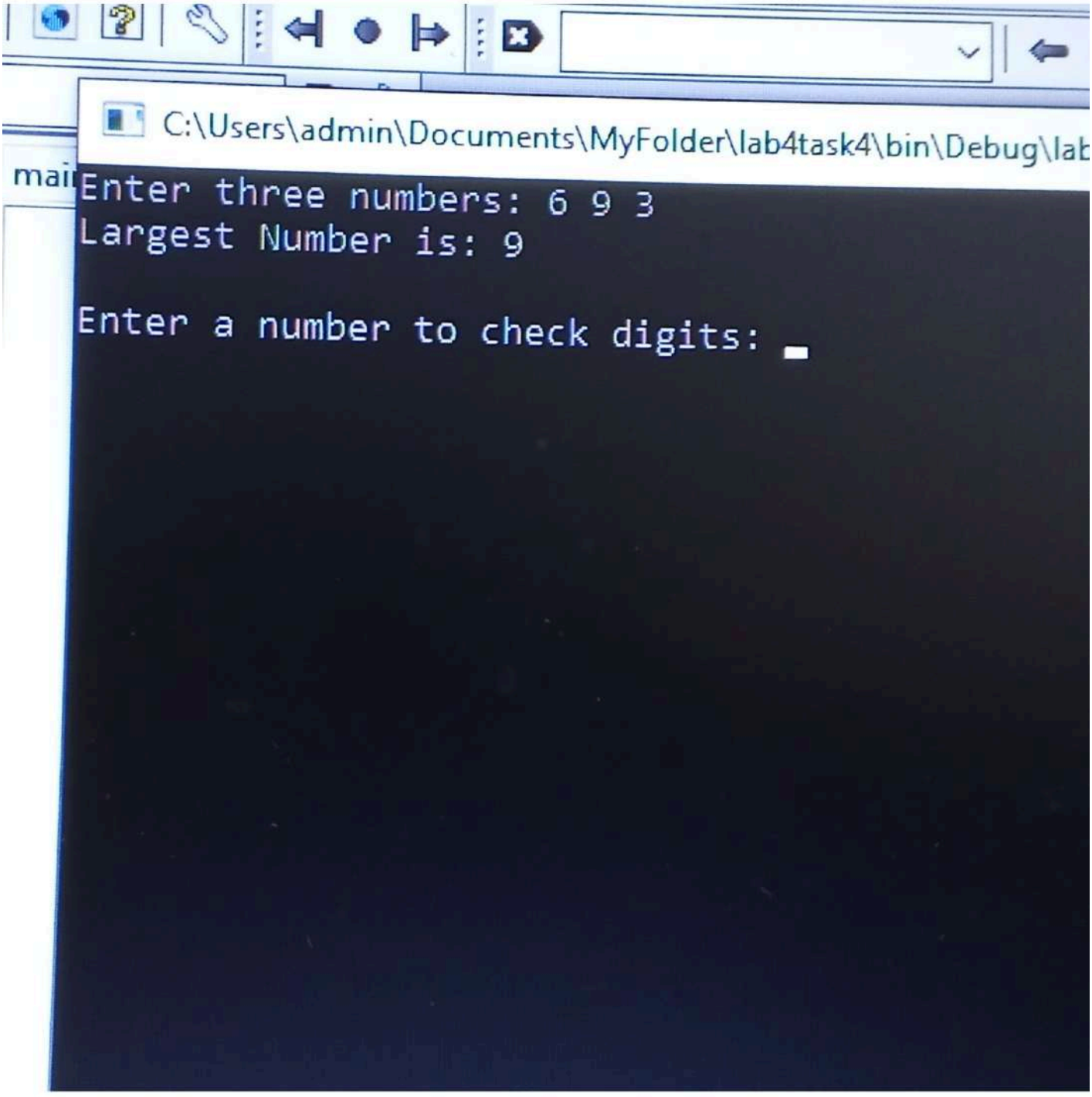
Enter three numbers: 6 9 3

Largest number is: 9


Debug
main.cpp
main.cpp
main.cpp
*main.cpp
1 #include <iostream>
2 using namespace std;
3 int main() {
4 int n1, n2, n3;
5 cout << "Enter three numbers: ";
6 cin >> n1 >> n2 >> n3;
7 int largest;
8 if (n1 >= n2) {
9 if (n1 >= n3) {
10 largest = n1;
11 } else {
12 largest = n3;
13 }
14 } else {
15 if (n2 >= n3) {
16 largest = n2;
17 } else {
18 largest = n3;
19 }
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messag


wxSmith
Tools
Tools+
Plugins
DoxyBlocks
Debug
main() : int
main.cpp
main.cpp
*main.cpp
} else {
if (n2 >= n3) {
largest = n2;
} else {
largest = n3;
}
}
cout << "Largest Number is: " << largest << endl;
// Part 2: Check if positive and count digits
int num;
cout << "\nEnter a number to check digits: ";
cin >> num;
if (num > 0) {
cout << "Number is Positive" << endl;
int count = 0;
int temp = num;


Debug
main.cpp
main.cpp
*main.cpp
cin >> num;
if (num > 0) {
cout << "Number is Positive" << endl;
int count = 0;
int temp = num;
while (temp != 0) {
temp = temp / 10;
count++;
}
cout << "Number of digits: " << count << endl;
} else {
cout << "Number is not Positive" << endl;
}
return 0;
}


C:\Users\admin\Documents\MyFolder\lab4task4\bin\Debug\lab
Enter three numbers: 6 9 3
Largest Number is: 9
Enter a number to check digits: